noc22_cs107

Mock Exam 2 | Course ID: noc22_cs107

Q1. According to the lecture, what is the 'software crisis'?

A) A shortage of hardware to run modern software applications.
B) A persistent set of problems including projects failing to meet requirements, exceeding budgets, and being delivered late.
C) The rapid increase in software costs relative to hardware costs.
D) The difficulty in finding developers trained in older programming languages.

Source: ilHPCbkZLV4 | Difficulty: Easy

Q2. What is a project stakeholder, as defined in the lecture?

A) Only the customers who pay for the software project.
B) Only the project manager and the development team.
C) Individuals or organizations actively involved in a project, or whose interests may be affected by its execution or completion.
D) The top management of the company that initiates the project.

Source: nmhuINAXksk | Difficulty: Easy

Q3. The '99 percent complete syndrome' refers to a situation where:

A) A project is successfully completed with 99% of its features.
B) A project is perpetually reported as 'almost done' by team members, leading to significant delays.
C) 99% of the project budget has been spent before completion.
D) The final product has a 99% defect-free rate.

Source: yC0A6_nYPYQ | Difficulty: Easy

Q4. What is the principle of 'phase containment of errors'?

A) Containing all project errors within a single, dedicated testing phase.
B) The idea that errors should be detected and corrected as close as possible to the phase in which they were introduced.
C) A method for ensuring that errors in one module do not affect other modules.
D) The principle that a certain number of errors is acceptable in each phase.

Source: J1-LpZr9GSQ | Difficulty: Easy

Q5. According to the lecture, what is a major criticism of the Waterfall Model?

A) It requires too much customer involvement throughout the project.
B) It is too flexible, leading to scope creep and chaos.
C) It is a 'heavy-weight process' with extensive documentation overhead and inflexibility to change requests.
D) It is only suitable for very small, trivial projects.

Source: 0O6fJSkafOo | Difficulty: Easy

Q6. What is the core philosophy of the evolutionary model of software development?

A) To define and freeze all requirements before development begins.
B) To develop core modules initially and then progressively refine and expand them through multiple iterations based on user feedback.
C) To complete the entire system in a single, large release after extensive testing.
D) To focus exclusively on risk mitigation through prototyping.

Source: sHoPCWz6YeQ | Difficulty: Easy

Q7. In the context of Extreme Programming (XP), what is 'pair programming'?

A) Two programmers dividing the work and coding separate modules in parallel.
B) A lead programmer mentoring a junior programmer.
C) Two programmers working together at one workstation, with one writing code and the other reviewing it.
D) A formal code review process conducted by a pair of senior developers.
Source: lWkKAZ0jatM | Difficulty: Easy

Q8. In the Scrum framework, who is the Product Owner?

A) The most senior developer on the team who owns the codebase.
B) The project manager responsible for the team's daily tasks.
C) A team member responsible for facilitating the Scrum process and removing impediments.
D) A team member who acts as the voice of the customer, defines features, and prioritizes the Product Backlog.
Source: 0JYkgfgKn8I | Difficulty: Easy

Q9. What is the 'Price to Win' estimation method?

A) A technique where the project cost is estimated based on the most competitive market price.
B) A method where the project cost is set at whatever the customer is willing or able to spend.
C) An estimation based on winning a government contract through the lowest possible bid.
D) A detailed bottom-up estimation to ensure the price is high enough to guarantee a win.
Source: wqyJF-Gbdp0 | Difficulty: Easy

Q10. What are the three development modes in the Basic COCOMO 81 model?

A) Simple, Moderate, Complex
B) Small, Medium, Large
C) Organic, Semi-detached, Embedded
D) Basic, Intermediate, Detailed
Source: sQyR78dlSzc | Difficulty: Easy

Q11. What is the primary purpose of a feasibility study in the project management lifecycle?

A) To create a detailed project schedule and allocate resources.
B) To determine if a project is technically viable and financially worthwhile before significant resources are committed.
C) To gather all customer requirements for the software product.
D) To select the development team and assign roles and responsibilities.
Source: rAo0iMok7lI | Difficulty: Medium

Q12. What is the key difference between project management processes and product-oriented processes?

A) Project management processes are done by the customer, while product-oriented processes are done by the development team.
B) Project management processes focus on planning and control, while product-oriented processes focus on creating the actual project deliverables.
C) Project management processes are only performed at the beginning of a project, while product-oriented processes are continuous.
D) There is no significant difference; the terms are used interchangeably.
Source: A8-PtgugnGE | Difficulty: Medium

Q13. What is the key difference between Project Risk and Business Risk as defined in the lecture?

A) Project Risk affects the budget, while Business Risk affects the schedule.

B) Project Risk relates to threats to the successful execution of the project itself, while Business Risk threatens the benefits of the delivered project.
C) Project Risk is managed by the project manager, while Business Risk is managed by the customer.
D) Project Risk is technical in nature, while Business Risk is purely financial.
Source: 8RuJfk1Xj2U | Difficulty: Medium

Q14. What is a key difference between the Payback Period and Net Present Value (NPV) as project evaluation techniques?

A) Payback Period considers overall profitability, while NPV only considers the time to recover investment.
B) Payback Period is more complex to calculate than NPV.
C) NPV accounts for the time value of money by discounting future cash flows, while Payback Period does not.
D) NPV is expressed in years, while Payback Period is a percentage.
Source: 8UlgpszPK6w | Difficulty: Medium

Q15. In the context of risk evaluation, how is a 'risk premium' used when calculating the Net Present Value (NPV) for a risky project?

A) It is a bonus paid to the project team for managing risks successfully.
B) It is an additional amount added to the project's expected benefits to account for potential gains.
C) It is an additional percentage added to the discount rate, making riskier projects appear less attractive unless their returns are higher.
D) It is a separate fund set aside to cover any financial losses from risks.
Source: GqMtULB0rLE | Difficulty: Medium

Q16. How does a Program Manager's role differ from a Project Manager's role regarding resource optimization?

A) A Program Manager minimizes resource demand for their specific project, while a Project Manager optimizes across all projects.
B) A Program Manager aims for the optimal use of resources across all projects within the program, while a Project Manager seeks to minimize resource demand for their single project.
C) A Program Manager focuses only on financial resources, while a Project Manager focuses on human resources.
D) There is no difference; both roles have the same objective for resource optimization.
Source: 5W5xe-CJZTQ | Difficulty: Medium

Q17. What is a primary disadvantage of the bottom-up estimation technique?

A) It is very fast to implement but often inaccurate.
B) It fails to consider the complexities of individual low-level tasks.
C) It can be very time-consuming and may overlook system-level costs like integration and configuration management.
D) It requires extensive historical data from similar past projects to be effective.
Source: U5v_uKQe0Ic | Difficulty: Medium

Q18. What is the purpose of the Value Adjustment Factor (VAF) in the Albrecht/IFPUG function point calculation?

A) To convert the final function point count into Lines of Code (LOC).
B) To determine the initial raw count of function points based on user inputs and outputs.
C) To refine the Unadjusted Function Point (UFP) count by accounting for 14 general system characteristics related to technical and environmental complexity.
D) To calculate the total cost of the project based on the function point size.
Source: MzhN6eZIUNU | Difficulty: Medium

Q19. COSMIC Full Function Points (FFPs) are designed for embedded and real-time systems. What are the four fundamental types of data movement that this method counts?

A) Inputs, Outputs, Inquiries, and Files
B) Creates, Reads, Updates, and Deletes
C) Entries, Exits, Reads, and Writes
D) Screens, Reports, Modules, and Interfaces

Source: rOn-GA71uKM | Difficulty: Medium

Q20. What is the purpose of calculating Program Volume (V) in Halstead's Software Science?

A) To measure the physical disk space occupied by the program.
B) To count the total number of operators and operands used in the program.
C) To compensate for the effect of the programming language used, providing a more standardized measure of program size in bits.
D) To estimate the number of mental discriminations required to write the program.

Source: w_Ka3XuXEbw | Difficulty: Medium

Q21. How does Putnam's model demonstrate the concept of 'schedule compression'?

A) It shows that reducing development time by half will also cut the required effort in half.
B) It indicates that compressing the schedule has a linear and predictable impact on project cost.
C) It shows a highly non-linear relationship where reducing development time leads to a disproportionately large increase in required effort (effort scales with td^4).
D) It proves that schedule compression is impossible beyond a 10% reduction from the nominal time.

Source: _KdYrzzMuBo | Difficulty: Medium

Q22. Why is it important for a project manager to avoid both too fine and too coarse granularity when creating a Work Breakdown Structure (WBS)?

A) Too fine granularity is expensive, while too coarse granularity is too simple.
B) Too fine granularity creates excessive management overhead, while too coarse granularity leads to a loss of project control and visibility.
C) Too fine granularity is only for agile projects, while too coarse granularity is for waterfall projects.
D) Too fine granularity requires expert developers, while too coarse granularity can be handled by junior developers.

Source: AvwZ7JO8LNk | Difficulty: Medium

Q23. How does the concept of 'slack' or 'float time' in PERT/CPM provide flexibility for a project manager in resource management?

A) It allows the project manager to add extra features to activities with slack time.
B) It indicates which activities can be eliminated from the project to save time.
C) It allows the project manager to strategically reallocate resources from non-critical activities (with slack) to critical activities that are falling behind.
D) It represents the extra budget available for non-critical activities.

Source: B6Awvb58gIA | Difficulty: Hard

Q24. During project crashing, why must the critical path be re-evaluated after each incremental reduction in a task's duration?

A) To ensure the project budget is not exceeded.
B) Because reducing the duration of a critical task can cause a different, previously non-critical path to become the new critical path.
C) To confirm that the customer still requires the project to be accelerated.
D) To allow the project manager to add more tasks to the project.

Source: ikOTZBPaklE | Difficulty: Hard

Q25. How does the Oldham-Hackman Job Characteristic Model suggest a manager can redesign a job to make it more meaningful and motivating?

A) By increasing the salary and improving working conditions.

B) By implementing stricter supervision and control over tasks.
C) By increasing skill variety, task identity, task significance, autonomy, and feedback.
D) By reducing the number of tasks to minimize employee workload.
Source: TQWJYqOzs0g | Difficulty: Hard

Q26. According to Boehm's list of frequent risks, what is 'gold plating' and how can it be mitigated?

A) It is the risk of using expensive hardware; it can be mitigated by choosing cheaper alternatives.
B) It is the risk of developers adding features not specified in requirements; it can be mitigated by requirement scrubbing and prototyping for user evaluation.
C) It is the risk of delivering the project late; it can be mitigated by adding more developers to the team.
D) It is the risk of the user interface being poorly designed; it can be mitigated by hiring a dedicated UI/UX expert.
Source: wjd0YMixlJA | Difficulty: Hard

Q27. The lecture advises not to add new dependencies to the activity network to reflect resource constraints. Why is this the recommended practice?

A) Because resource constraints are not as important as task dependencies.
B) Because adding such dependencies makes the network messy, and if the resource constraint disappears, the artificial link remains, causing confusion.
C) Because modern software tools cannot handle resource-based dependencies.
D) Because it violates the fundamental rules of PERT/CPM network construction.
Source: ZcO5UNzMtz8 | Difficulty: Hard

Q28. The lecture mentions that Earned Value Analysis (EVA) has not achieved universal acceptance in software development. What is the primary reason for this challenge?

A) EVA calculations are too complex for most project managers to perform.
B) EVA is only suitable for hardware projects and cannot be applied to intangible products.
C) A 'half-completed software project' often has no functional value until it is fully deployed, making it difficult to assign meaningful 'earned value' to partial work.
D) The US Department of Defense has prohibited the use of EVA for software projects.
Source: I81fOjyzb0E | Difficulty: Hard

Q29. What are the potential consequences and risks of altering activity precedence requirements to get a project back on target?

A) It always reduces project costs and improves team morale.
B) It has no impact on the project other than accelerating the schedule.
C) It can compromise quality and increase project risk, such as starting coding before design is complete, which can lead to extensive rework.
D) It is a forbidden practice in all project management methodologies.
Source: Ar0ynpJ86QU | Difficulty: Hard

Q30. The lecture states that an organization cannot advertise its products as 'ISO 9000 certified.' Why is this the case, and what is the underlying assumption of ISO 9000 certification?

A) Because ISO 9000 is an outdated standard that has been replaced by SEI CMM.
B) Because advertising individual products is too expensive for most ISO certified companies.
C) Because ISO certification is awarded to an organization's PROCESS, not its products, based on the assumption that a good process leads to a good product.
D) Because only the ISO organization itself is allowed to advertise certified products, not the companies that make them.
Source: ODQNex8xtjU | Difficulty: Hard

Answer Key

1: B	2: C	3: B	4: B	5: C
6: B	7: C	8: D	9: B	10: C
11: B	12: B	13: B	14: C	15: C
16: B	17: C	18: C	19: C	20: C
21: C	22: B	23: C	24: B	25: C
26: B	27: B	28: C	29: C	30: C